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Flagship · Computer Vision & Healthcare AI

BwellCheck · Contactless Physiological Monitoring

Extracting clinical-grade heart rate, respiration, and HRV telemetry from standard camera feeds using rPPG and facial ROI tracking · Moving through CDSCO regulatory clearance.

KEY METRICS

~25-30%
SIGNAL STABILITY GAIN

CDSCO
REGULATORY PATHWAY

rPPG
REMOTE PHOTOPLETHYSMOGRAPHY

1. Context · Clinical Telemetry without Wearables

Cardiovascular telemetry usually requires contact sensors - ECG leads, pulse oximeters, or specialized hardware. BwellCheck solves this by turning any standard RGB webcam or smartphone camera into a clinical-grade physiological sensor through remote photoplethysmography (rPPG).

By measuring micro-variations in skin reflectance caused by subcutaneous blood volume pulses, the system extracts real-time heart rate, respiratory rates, and heart rate variability (HRV) without any physical patient contact.

2. Systems Engineered · Signal Processing & ML

- **rPPG Signal Extraction Pipeline:** Implemented CHROM (Chrominance-based) and POS (Plane-Orthogonal-to-Skin) algorithms in Python to isolate pulsatile blood volume signals from noisy RGB video streams.
- **Spectral Analysis & Filtering:** Designed Fast Fourier Transform (FFT) and Butterworth respiratory bandpass filters to calculate heart rate, respiration rate, and HRV metrics (RMSSD, LF/HF power spectral ratios).
- **Facial Landmark ROI Tracking:** Integrated MediaPipe FaceMesh to dynamically anchor regions of interest across forehead and malar zones.
- **Predictive Proxy Modeling:** Trained and integrated Random Forest regressors to estimate physiological proxy trends (blood pressure and blood glucose indicators) from extracted wave features.

+28%

Signal Stability Improvement

Dynamic MediaPipe FaceMesh ROI tracking anchors facial micro-regions across ±25° head movement, reducing motion artifacts over raw bounding box extraction.

REGULATORY NOTE · CDSCO COMPLIANCE

CDSCO Medical Device Regulatory Pathway

Navigated the formal CDSCO regulatory pathway covering Health Technology Assessment (HTA), Quality Management Systems (QMS), and software and clinical evaluation test licences. Attended government regulatory sessions and converted the compliance notes into an actionable engineering framework for our clinical trial pipelines.

3. Outcome · Validation & Impact

BwellCheck transitioned contactless vitals monitoring from an unstable prototype into a validated clinical pipeline. The enhanced FaceMesh ROI tracking and frequency filtering deliver consistent signal fidelity under real-world ambient conditions.

The platform is actively navigating CDSCO test licencing, providing BWell HealthTech with a clear regulatory moat for contactless health telemetry.

> *Most AI engineers build models. Few navigate the regulatory pathway that turns a model into a product patients can actually use.*

TECHNOLOGIES & TOOLING

Python · rPPG · MediaPipe FaceMesh · FFT Signal Analysis · Random Forest · FastAPI · WebSockets · React.js / Next.js

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